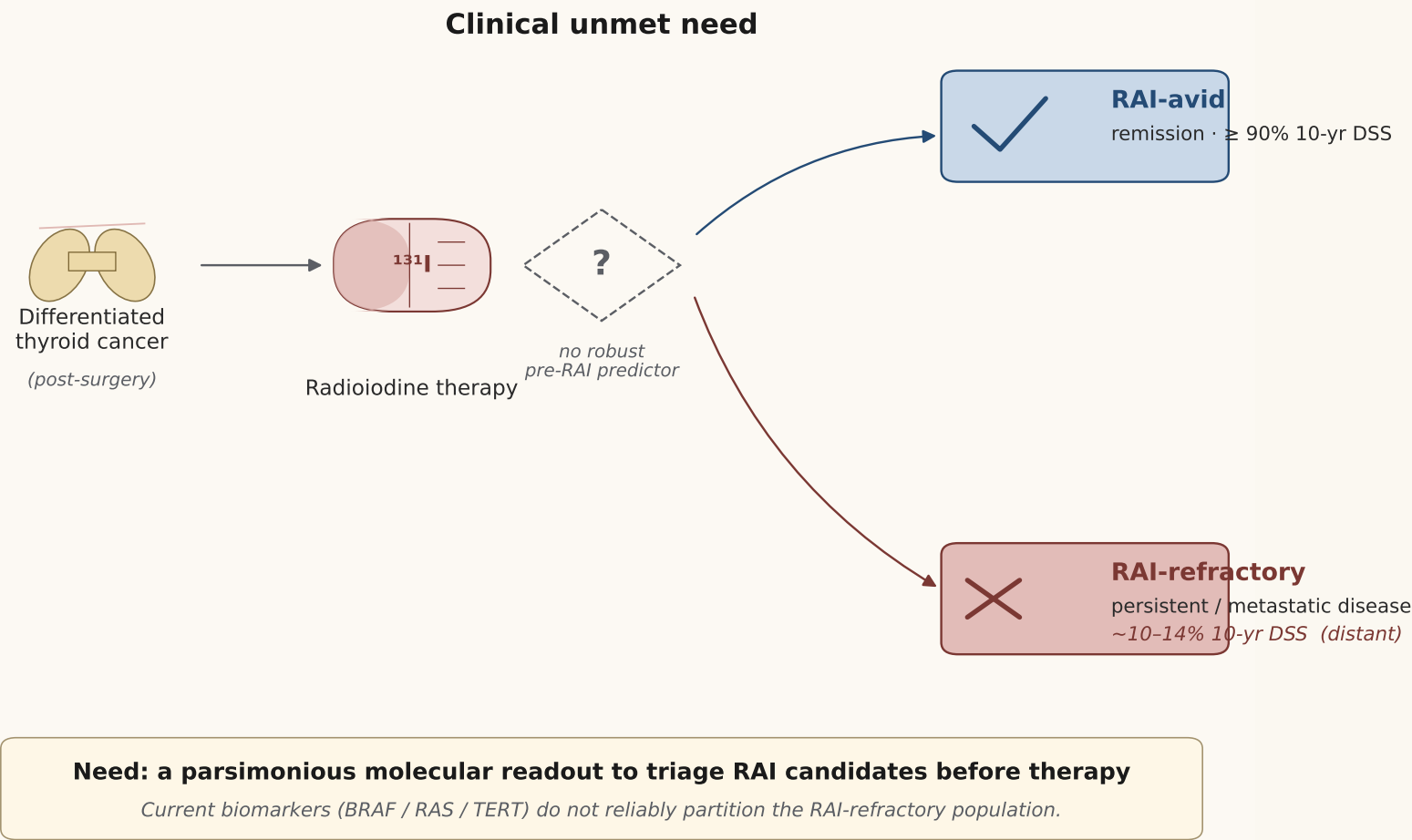
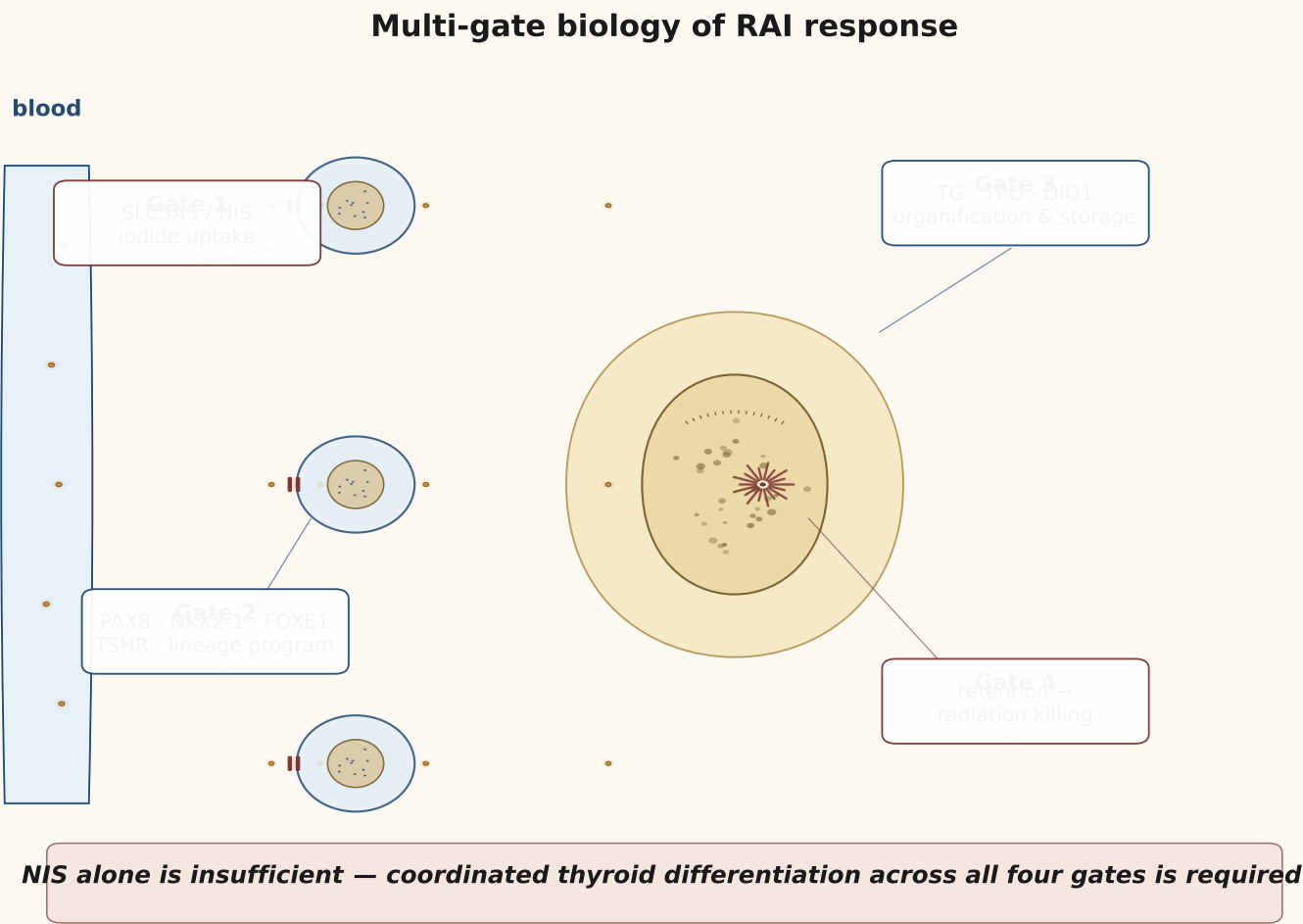


A multi-gate model of radioiodine failure in thyroid cancer

a

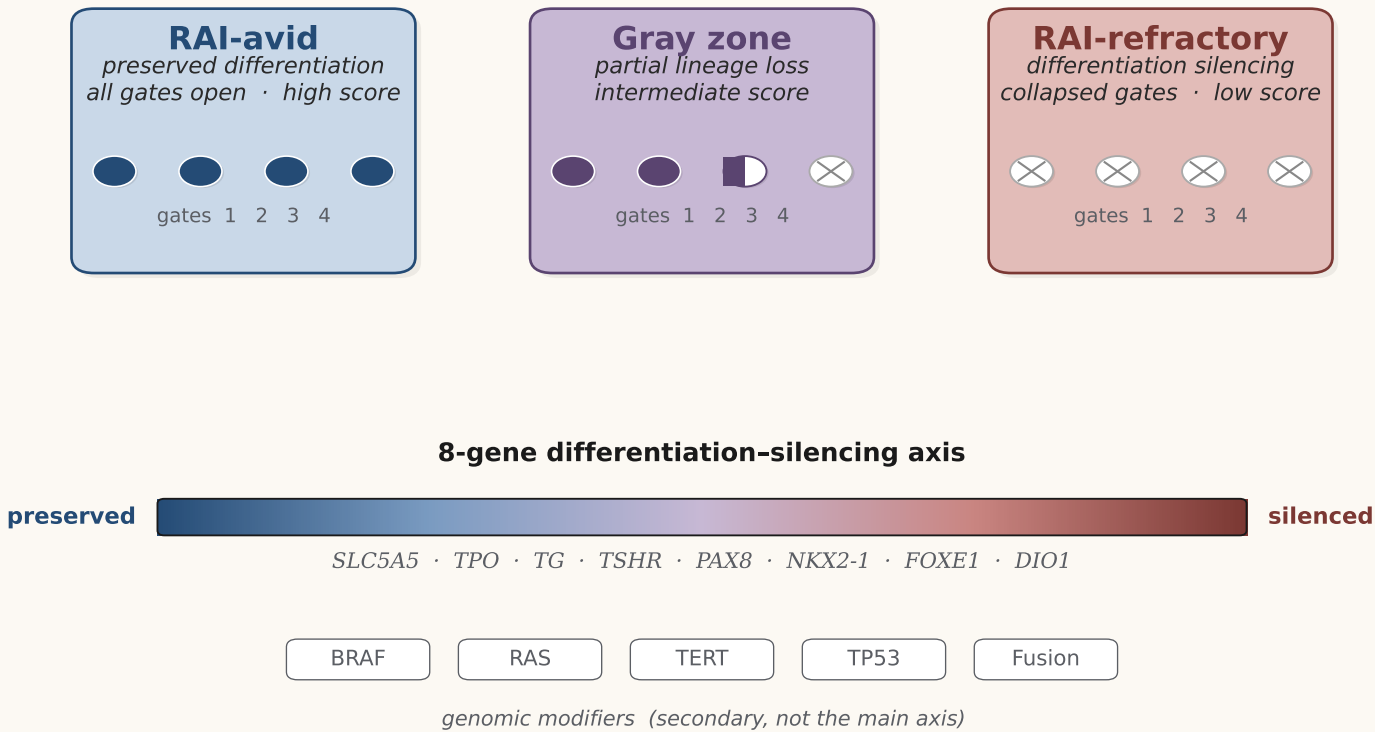


b



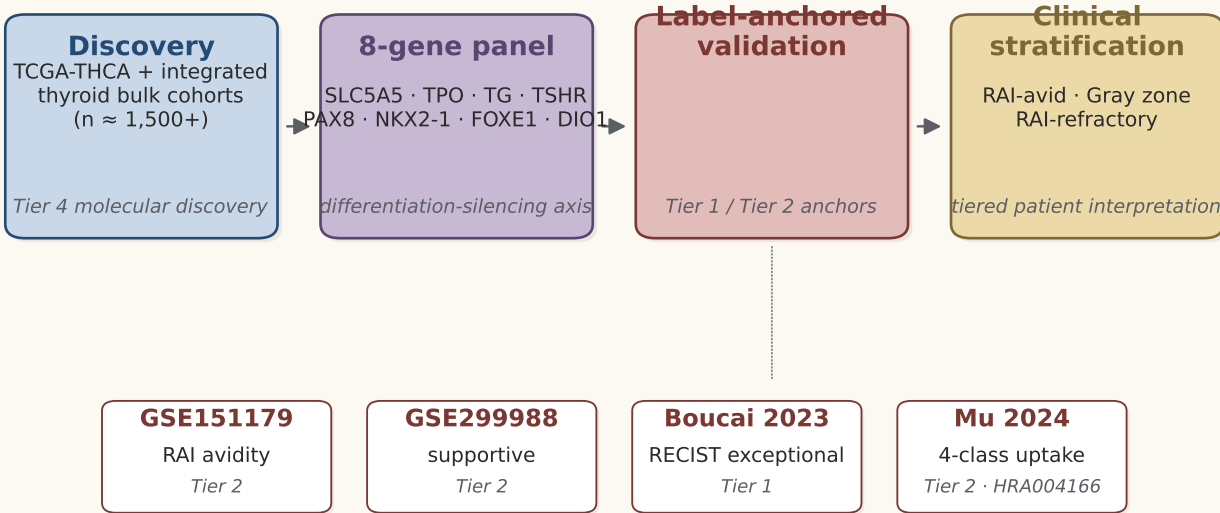
c

Three molecular states along the eight-gene differentiation-silencing axis



d

Tiered evidence ladder · discovery → validation → stratification



Tier 4 discovery → parsimonious 8-gene panel → label-anchored validation → three-class stratification

Risk-stratification readout, not a stand-alone clinical biomarker— prospective Tier-1 confirmation is the next step.